FAMA–Band socket hub (rigid–flex) – hierarchical root. All inter–sheet nets are global labels.

Power

File: power.kicad_sch

Power: USB–C, charger, battery, 3V3 buck–boost, AFE supply + USB interlock

Analog front end

File: afe.kicad_sch

Analog front end: ADS1298, 2.048 MHz clock, input filters, electrode connectors

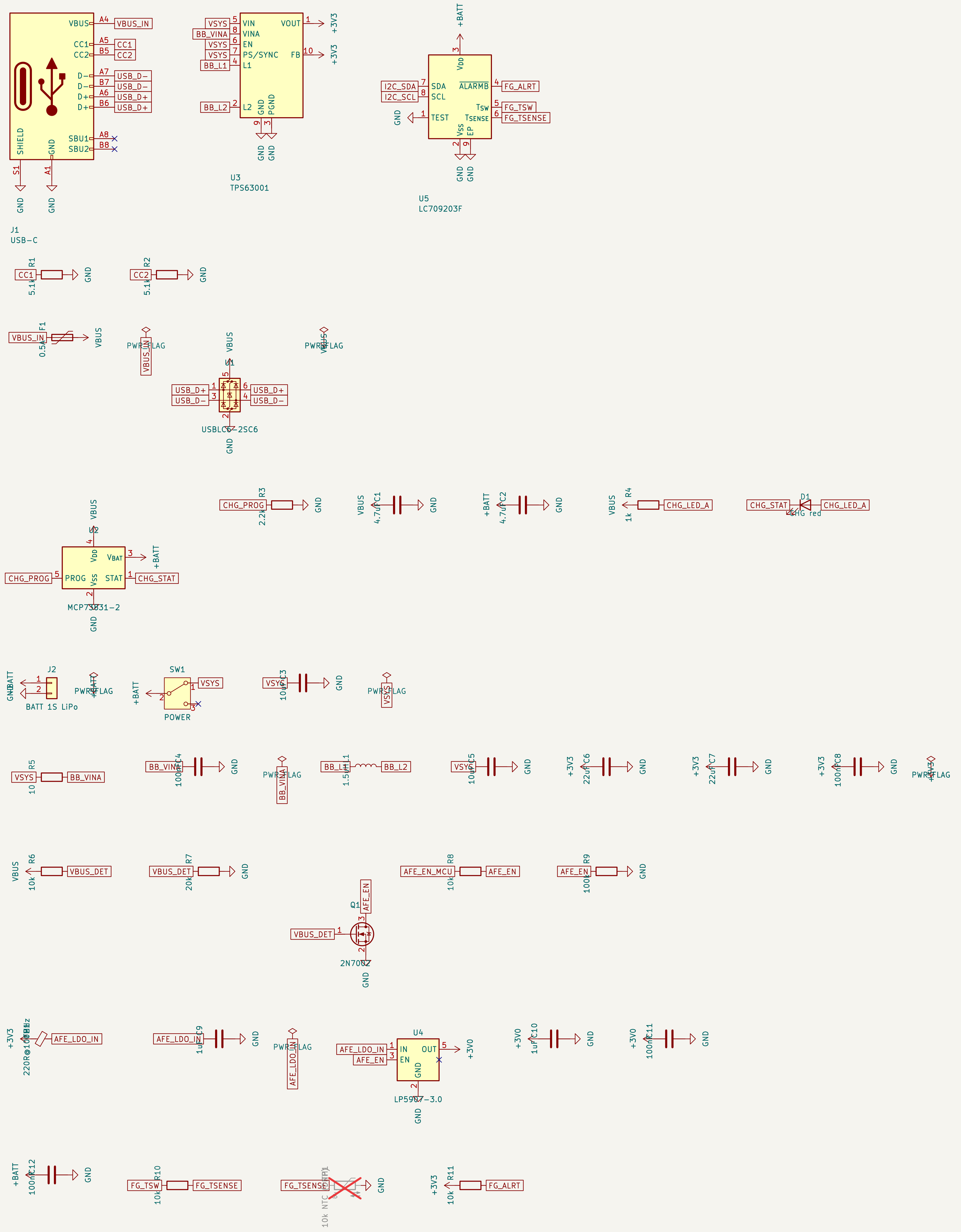
MCU

File: mcu.kicad_sch

MCU: ESP32–S3 module, IMU, microSD, buttons, LEDs

Generated by hardware/gen/gen_sch.py from the Python netlist – edit the netlist, not this file
FAMA–Net thesis hardware
Sheet: /
File: hub_flex.kicad_sch
Title: FAMA–Band socket hub (rigid–flex) – Power
Size: A3 Date: 2026–10–25 **Rev: A**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4 Id: 1/4

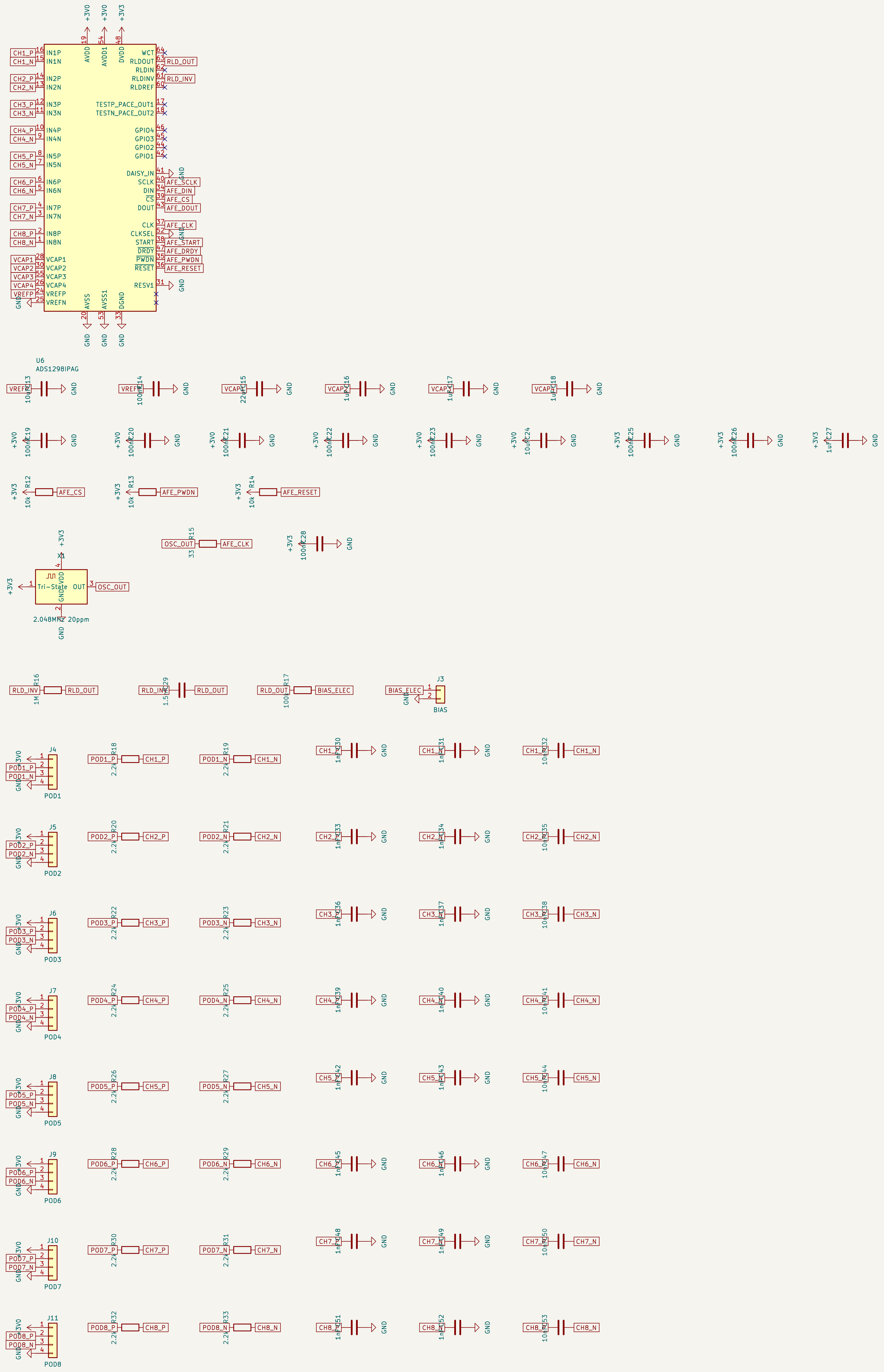
Power: USB–C, charger, battery, 3V3 buck–boost, AFE supply + USB interlock



USB–C is for charging, firmware and data offload ONLY while the band is OFF the arm.
Q1 holds the AFE LDO in shutdown whenever VBUS is present, independent of firmware,
so electrodes are never energised while a mains–referenced USB host is attached.

Generated by hardware/gen/gen_sch.py from the Python netlist – edit the netlist, not this file
FAMA–Net thesis hardware
Sheet: /Power/
File: power.kicad_sch
Title: FAMA–Band socket hub (rigid–flex) – Power
Size: A2 Date: 2026–10–25 **Rev: A**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4 Id: 2/4

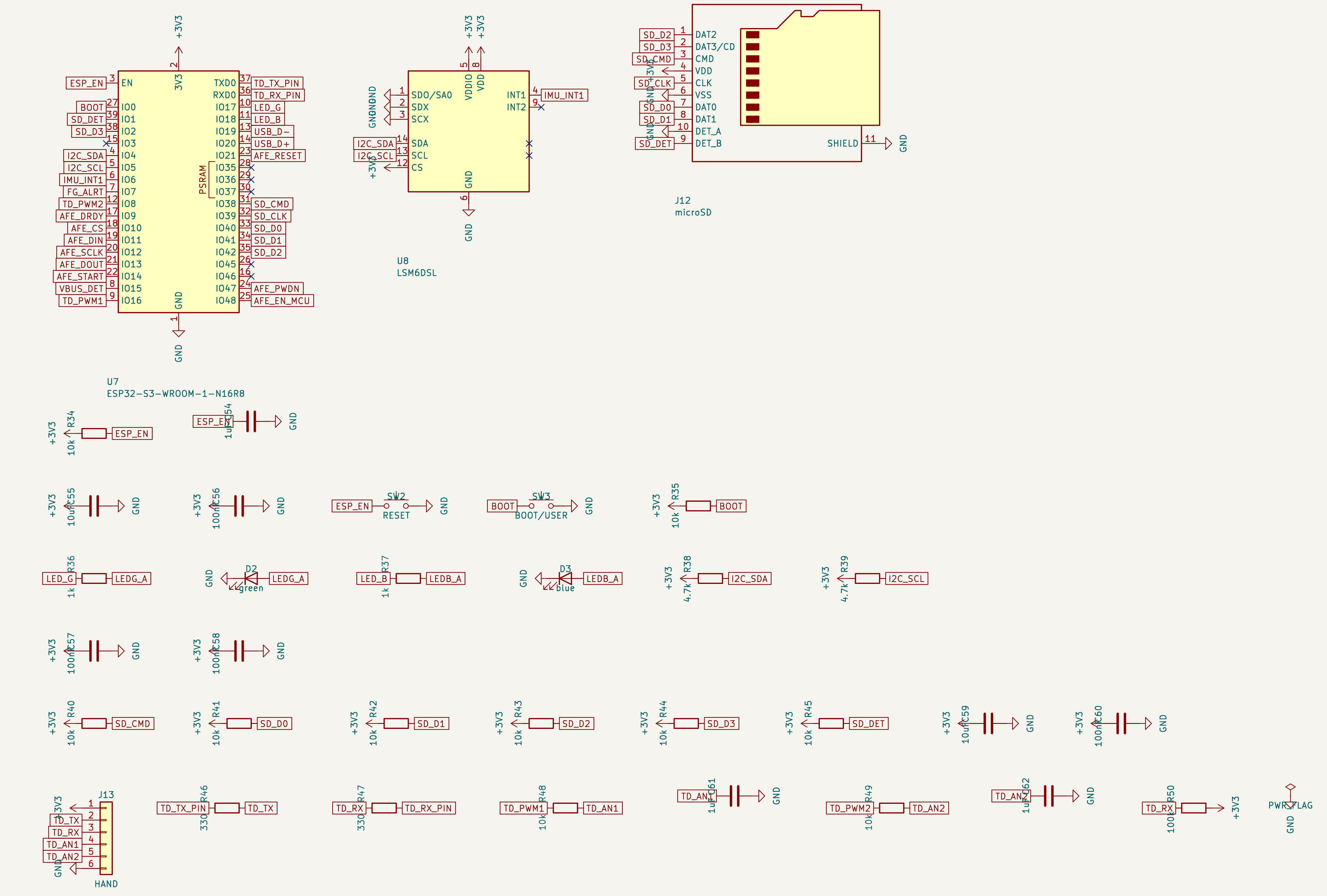
Analog front end: ADS1298, 2.048 MHz clock, input filters, electrode connectors



ADS1298 runs unipolar: AVDD = +3V0 (LP5907), DVDD = +3V3, VREF = 2.4 V internal.
Clock: external 2.048 MHz +/−20 ppm (CLKSEL = 0) −> f_s = 1000.00 Hz exactly (DR = 101, HR mode).
Channel n: pod buffer −> 2.2k + 1n (CM) / 10n (DM) EMI filter −> INnP/INnN, PGA gain 6,
range +/−400 mV, 1 LSB = 47.7 nV. Right−leg drive: RLDOUT −> 100k −> bias electrode.

Generated by hardware/gen/gen_sch.py from the Python netlist − edit the netlist, not this file
FAMA−Net thesis hardware
Sheet: /Analog front end/
File: afe.kicad_sch
Title: FAMA−Band socket hub (rigid−flex) − Analog front end
Size: A1 Date: 2026−10−25 **Rev: A**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4 Id: 3/4

MCU: ESP32-S3 module, IMU, microSD, buttons, LEDs



Terminal-device (prosthetic hand) interface: UART 115200 8N1 result packets + two
filtered PWM proportional outputs (fc = 16 Hz). Outputs are disabled until the
prosthesis enables them (firmware mode). Hand powered from its own battery.

ESP32-S3-WROOM-1-N16R8 (octal PSRAM: GPIO35-37 reserved). Strapping pins 0/3/45/46 left at defaults.
AFE on SPI2 (GPIO9-14,21,47,48), microSD 4-bit SDMMC (GPIO1,2,38-42), I2C IMU + fuel gauge (GPIO4/5).

Generated by hardware/gen/gen_sch.py from the Python netlist – edit the netlist, not this file
FAMA-Net thesis hardware
Sheet: /MCU/
File: mcu.kicad_sch
Title: FAMA-Band socket hub (rigid-flex) – MCU
Size: A3 Date: 2026-10-25 **Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 4/4